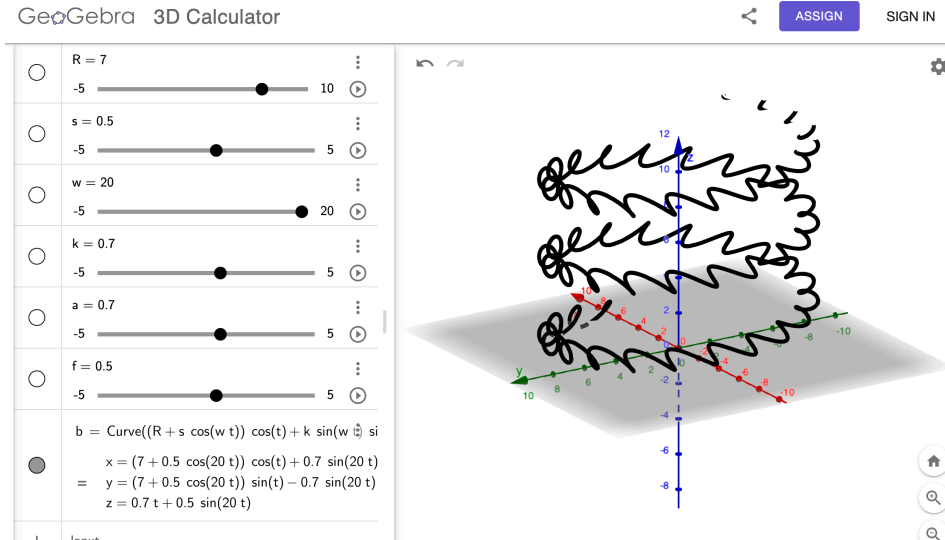


This is an essay I wrote as a part of my research project on modelling the Earth's trajectory. For more compact viewing, I included the resulting model function, scaled for clearer viewing in this preface. It is the image showed below. The rest of the essay covers how I got to this result explaining the shapes, magnitudes and scaling steps in more detail.



Preface Image: Model of Earth's Trajectory Scaled for Clearer Viewing of the Path

The function presents 3 key movements (in order from more zoomed in to more general):

1. Earth's orbit around the Sun (the small, tilted loops)
2. Movement of the Sun through the Milky way (the circle that the small loops travel in)
3. The movement of Milky Way through the Galaxy (Slow vertically upward progression of the small loops in a loop, forming a double helix on graph 11)

It also highlights the following main features characteristic to the Earth's path:

1. Tilt between the orbital planes of the Earth around the Sun and the Sun in Milky Way
2. The Earth's characteristic elliptical orbit
3. The Sun's approximated elliptical orbit

The final resulting scalable parametric function that captures all these movements is:

$$c(t) = \text{Curve}((R + s \cos(w t))\cos(t) + k \sin(w t)\sin(t), (R + s \cos(w t))\sin(t) - k \sin(w t)\cos(t), at + f \sin(wt), t, 0, 2\pi)$$

Modelling Earth's Trajectory Through the Universe

Introduction

Thinking about the YouTube video “How fast are you moving through the universe?” by Seeker (channel) inspired my curiosity to think about the trajectory of the Earth. Being very interested in astrophysics and finding this topic particularly interesting to think about, I decided to use the opportunity of the Maths IA to accurately model this movement. The aim of the investigation is to settle my curiosity about the Earth's motion and astrophysics by modelling the Earth's trajectory as accurately as I can. To model this, cylindrical coordinates will be used. They are a three-dimensional extension of polar coordinates which combine the polar coordinate system of the xy plane with a cartesian form of the z coordinate. I will use GeoGebra's 3D calculator.

I concluded that the trajectory must be a helix shape (caused by the Sun's movement and the Earth orbiting it) inside of a helix shape which is caused by the Sun's orbit around Milky Way and the Milky Way's movement through space. However, the z -axis stretch of the helix representing the Sun's orbit around Milky Way depends on the relative velocity of the Milky Way. The issue with this is that in the universe, there isn't a zero-velocity point meaning that the velocity of the Milky Way can be interpreted differently from an infinite number of velocities each of which are equally valid as a zero-velocity reference point. Keeping this in mind, the final function for this IA will instead be presented as a helix around circle shape left with a slider which can be adjusted based on different reference points.

Important Data

Quantity	Value
Perihelion ¹	$1.47098074 \times 10^{11}m$ ($1.47 \times 10^{11}m$)
Aphelion ²	$1.52097701 \times 10^{11}m$ ($1.52 \times 10^{11}m$)
Ecliptic plane eccentricity ³	0.01671123
Angle between ecliptic plane and galactic plane ⁴	60.3°
Duration of Earth's orbit	365.25 Days
Velocity of Solar system ⁵	720000 km/h
Distance from the Sun to the center of Milky Way ⁶	26000 light years

Table 1: Required Values

¹ (Espenak 2021)

² ibid

³ (NASA n.d.)

⁴ (Qmechanic 2019)

⁵ (NASA 2021)

⁶ (Schawinski 2015)

Perihelion – the point along the Earth’s orbit where its distance from the Sun is the shortest

Aphelion - the point along the Earth’s orbit where its distance from the Sun is the longest

Ecliptic plane – imaginary plane enclosed by the Earth orbit around the Sun (ellipse)

Galactic plane – imaginary plane enclosed by the Sun’s orbit around Milky Way

Cylindrical Coordinates

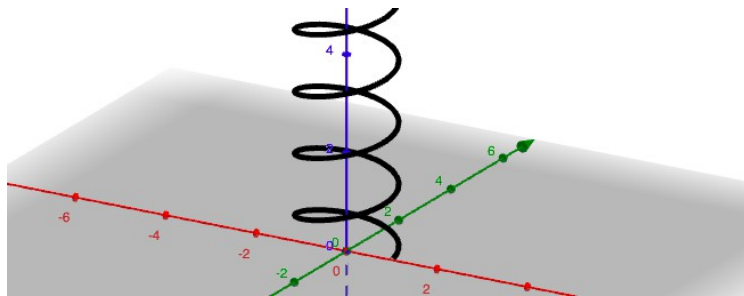
The system for denoting cylindrical coordinates is the following:

$c = \text{Curve}(\text{Expression}, \text{Expression}, \text{Expression}, \text{Parameter}, \text{Start Value}, \text{End Value})$

Where the expressions for x , y and z respectively are given as their parametric equations. The start and end values allow us to set the domain of the parameter. Since the model developed in this IA will be based on a helix, the basic function for a helix in cylindrical coordinates is given by curve 1:

$$c_1 = \text{Curve}(\cos(t), \sin(t), a t, t, 0, 100)$$

Making x vary as a function of cosine and y vary as a function of sine makes the xy variation of the curve circular as the x and y vary as the unit circle. z being a linear function of t makes the unit circle shape progress around the z -axis continuously making the curve into a helix.



Graph 1: Basic Helix Shape (curve 1)

Graph 1 plots the first curve which represents the movement of the Sun around Milky Way if we imagine the Milky Way’s center’s path as the z -axis. The Earth’s trajectory around this curve will be added as a helix going around this helix. It will be added later in this IA using more complex transformations. “ a ” is the undetermined stretch that we are free to choose because of there not being a zero-velocity reference point in the universe. Therefore, the Milky Way’s movement can be viewed in infinitely many ways and the stretch for “ a ” will be left as a slider.

Calculating Required Values

The Radius of the Sun's Orbit around Milky Way

Due the lack of accurate data about the trajectory of the Sun's orbit or other values which could be used to determine it, the Sun's orbit will be assumed as circular. It is difficult for scientists to determine these parameters because in Kepler's laws which are used to explain orbits, the bodies being orbited around are treated as point masses which is a fair assumption considering that the radii of planets and stars are of negligible lengths when compared to the radii of their orbits. This is certainly not the case with the Sun's orbit around Milky Way because the Sun is inside the Milky Way and thus the radius R of Milky Way is not a negligible length compared to the radius of the Sun's orbit. Furthermore, Milky Way's mass does not vary uniformly and the Sun throughout its orbit often enters areas of higher and lower densities such as star clusters which therefore affect its trajectory. The approximate distance from the Sun to the center of Milky Way is 26000 light years and that is the orbit radius used. A light year is the distance light travels in a year. To convert into meters, it is multiplied by the number of seconds in a year and then by the speed of light in $\frac{m}{s}$:

$$R = (26000 \times 365.25 \times 24 \times 60 \times 60)s \times 300\,000\,000 \frac{m}{s} = 2.5 \times 10^{20}m$$

Number of Earth's Orbits around the Sun in one Sun's Orbit around Milky Way

This number is necessary for modelling the function due to the nature of the transformations which will be made. Since the velocity of the Sun (equals the velocity of Solar system from table 1) is $720000 \frac{km}{h}$, converting to meters per second:

$$720000 \frac{km}{h} \times \frac{1000}{3600} = 200000 \frac{m}{s}$$

The circumference of the Sun's orbit is

$$C = 2\pi r = 2 \times \pi \times 2.5 \times 10^{20}m = 1.5 \times 10^{21}m$$

With length and velocity, using the rearranged equation for velocity ($t = \frac{C}{v}$), we can calculate the time required for the Sun to complete one orbit around Milky Way:

$$t = \frac{1.5 \times 10^{21}s}{200000 \frac{m}{s}} = 7.7 \times 10^{15}s$$

Converting to years (unit based on the Earth's orbit around the Sun), we get

$$t = \frac{7.7 \times 10^{15}s}{60 \times 60 \times 24 \times 365} = 240000000 \text{ years}$$

Equating to 240000000 Earth's orbits per one Sun's orbit

The Semi-minor Axis and Semi-major Axis of the Ecliptic Plane

The Earth's orbit helix will be made elliptical by adding stretches in the xy plane different from the stretches in the z direction on the 3-dimensional graph of the Earth's orbit. Therefore, the stretches applied in both directions need to be calculated using data from table 1.

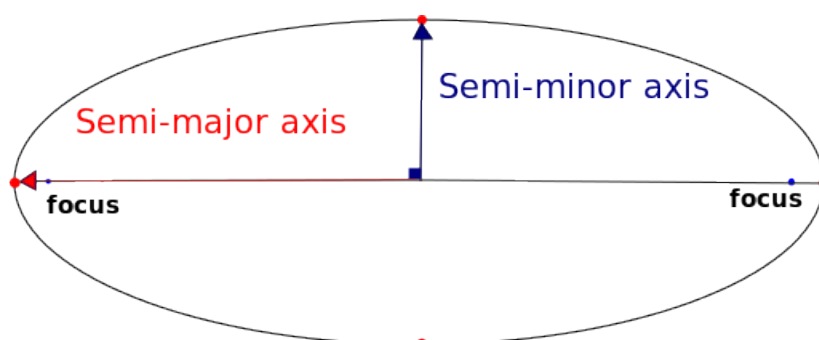


Diagram 1: Ellipse⁷

According to Kepler's first law studied in IB Physics, a planet's orbit around the Sun is always an ellipse and the Sun is located at one focus. Understanding that Perihelion and Aphelion are respectively the shortest and longest distance in the Earth's orbit from the Sun, and that the Earth's location at those distances are at the two points of intersection of the Major axis with the ellipse, the following formula can be derived:

$$2s = s_{per} + s_{aph}$$

Where s_{per} and s_{aph} are the Earth's distance from the Sun at the Perihelion and Aphelion respectively and s is the semi-major axis. Using table 1 values for s_{per} and s_{aph} , we get:

$$s = \frac{1.47098074 \times 10^{11} m + 1.52097701 \times 10^{11} m}{2} \Rightarrow s = 1.49597888 \times 10^{11} m$$

Understanding that $eccentricity = \frac{\text{distance from the center of the ellipse to the focus}}{\text{length of the semi-major axis}}$, where

eccentricity is ε , distance from the center to the focus is s_{ctf} , the following formula applies⁸:

$$\varepsilon = \frac{s_{ctf}}{s}$$

Rearranging and calculating:

$$s_{ctf} = \varepsilon s \Rightarrow s_{ctf} = 0.01671123 \times 1.49597888 \times 10^{11} m$$

$$\Rightarrow s_{ctf} = 2.499965 \times 10^9 m$$

To find the semi-minor axis b , the following rule for ellipses is used⁹:

⁷ (File:An image describing the semi-major and semi-minor axis of ellipse 2014)

⁸ (Eccentricity of the Ellipse n.d.)

$$s_{ctf}^2 = s^2 - b^2$$

Rearranging and calculating gives:

$$b^2 = s^2 - s_{ctf}^2 \quad \Rightarrow \quad b = \sqrt{s^2 - s_{ctf}^2}$$

$$\Rightarrow \quad b = \sqrt{(1.49597888 \times 10^{11}m)^2 - (2.499965 \times 10^9m)^2} \quad \Rightarrow \quad b = 1.495770 \times 10^{11}m$$

This means that the correct elliptical orbit of the Earth has $b = 1.495770 \times 10^{11}m$ and $s = 1.49597888 \times 10^{11}m$ as factors for its stretch in the xy plane and the z direction.

Creating the Circular Helix Curve

Modelling the 2D Plane of the Earth's Trajectory around Milky Way

To make complex transformations for the base helix-around-helix curve simpler, the xy plane will first be adjusted separately from the z -axis. Imagining how the Earth's trajectory looks in 2 dimensions on the galactic plane (xy plane), we get the following (not to scale):

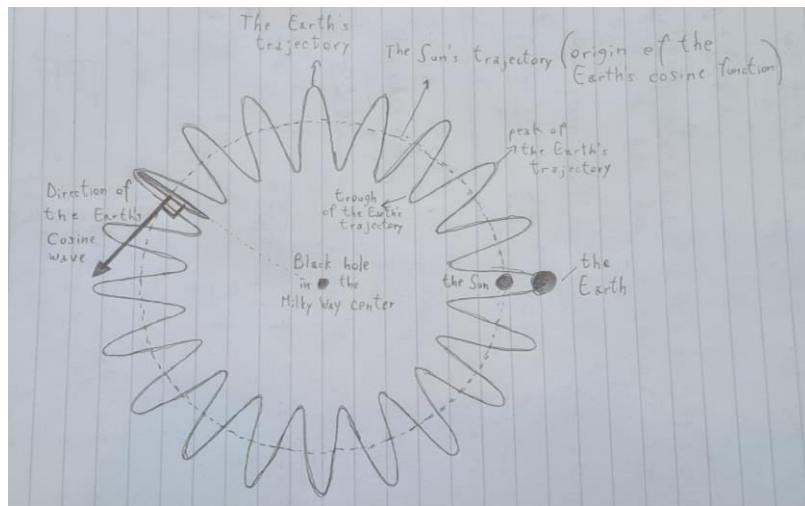


Diagram 2: 2-dimensional Trajectory of the Earth around Milky Way

As seen, the trajectory is the shape of a circular sine/cosine wave. For this investigation, the cosine wave will be chosen but both are equally correct. The direction of the Earth's cosine wave (normal to the ecliptic plane) is always perpendicular to the radius of the Sun's orbit. This can be logically concluded because the gravity of the black hole (in the Milky Way's center) cannot impact some masses and some not. If it is pulling the Sun, it also has to pull the Earth. Therefore, the normal to the ecliptic plane has to keep changing direction. To create

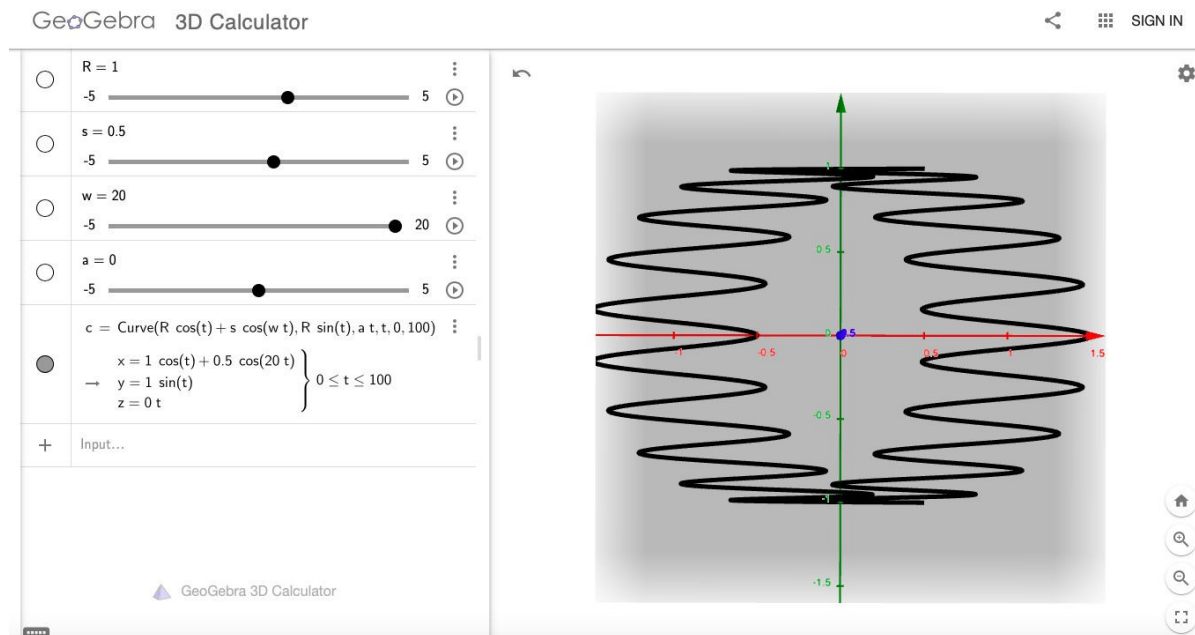
⁹ ibid

this curve, transformations are applied to curve 1. The added R is the radius of the Sun's orbit around Milky Way. (x and y are given the same stretch because due to the lack of accurate data of the orbit eccentricity, the orbit is assumed as circular):

$$c_2 = \text{Curve}(R \cos(t), R \sin(t), at, t, 0, 100)$$

To make this curve vary as a cosine function as on diagram 2 with respect to the x -axis, we need to shift the curve along the x -axis by amounts which vary as a cosine function throughout the curve. Therefore, the added shifts to the x -axis must oscillate throughout the curve. To achieve this, we add $\cos(w t)$ to the equation for x . Doing so causes the x -axis values of points on the curve to shift by the output of $\cos(w t)$ at that point. If, for example $w = 20$, this means that the value of $\cos(w t)$ will complete 20 oscillations while the original curve completes 1 circle. Thus, the shift in the x -axis oscillates 20 times throughout the main circle (the Sun's trajectory). Therefore, different parts of the curve shift 20 times in the x -axis by every value in the range of the function $+\cos(w t)$. Factor s is added to vary the magnitude of the shift. The new curve is:

$$c_3 = \text{Curve}(R \cos(t) + s \cos(w t), R \sin(t), at, t, 0, 100)$$

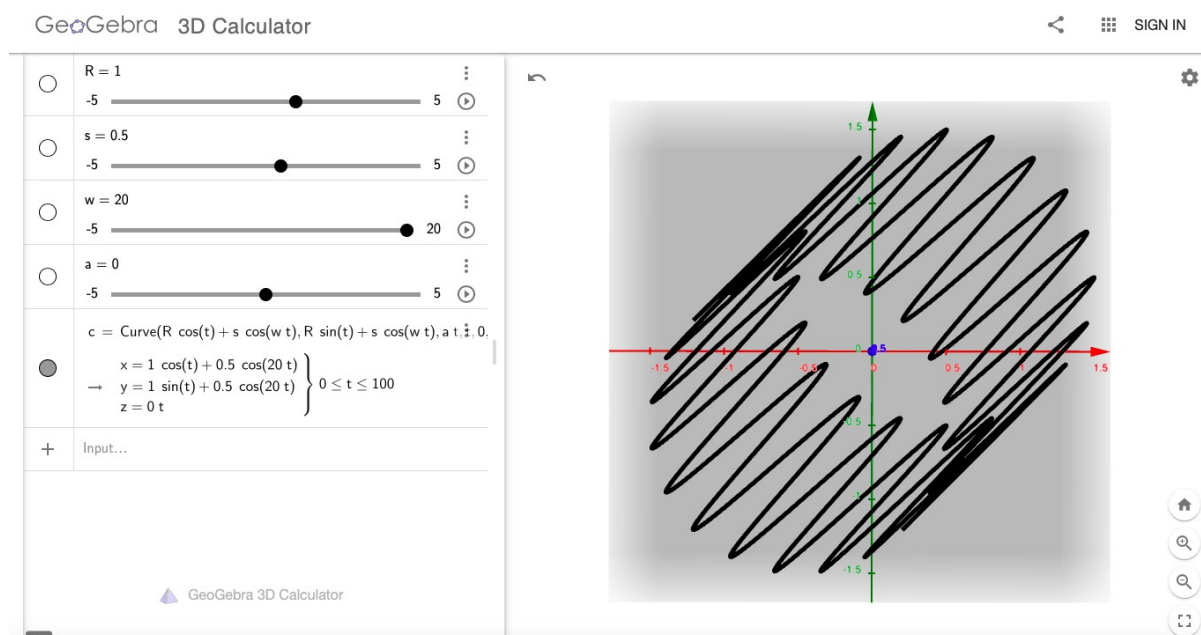


Graph 2: Cosine Transformation in the x -direction (curve 3)

Since we are making the cosine wave go in a circle, we refer back to diagram 1 to look at the cosine wave variation in the first quadrant of the orbit. We notice that whenever there is a trough, there is a shift in the values along the x -axis as well as along the y -axis towards the negative direction. Therefore, to make our curve vary more like the curve on diagram 1, the y -axis shifts have to match the x -axis shifts in terms of direction in the first quadrant. The y -axis

has to vary the same way as the x-axis in the first quadrant. (The shift in other quadrants will be adjusted in the following transformation). Therefore, to match the x-axis shift, the same $+ \cos(w t)$ transformation is added to the equation for y. The trough of $y + \cos(w t)$ which causes a negative shift on the y axis will happen at the same place where the trough for $x = + \cos(w t)$ happens and this will be the trough of the circular cosine wave (the part of the Earth's orbit where the Earth is the closes to the center of Milky Way). Because for easier transformations, the xy plane is set as one direction of stretch for the ecliptic plane ellipse and the z direction as the other direction of stretch, the same factor s is added to alter the magnitude of stretch on the y-axis. This gives:

$$c_4 = \text{Curve}(R \cos(t) + s \cos(w t), R \sin(t) + s \cos(w t), at, t, 0, 100)$$



Graph 3: Cosine Oscillations Around the Curve (curve 4)

However, as explained, because the gravity of the Milky Way has to affect the Earth also, the direction of the Earth's cosine wave has to change whereas on graph 3 it is constant. This needs to be changed. On diagram 1, we notice that the extent to which the trough of the circular cosine wave is indented towards the center of the orbit needs to be constant throughout the curve to maintain the same orbit of Earth around the Sun. This means that the sum of x and y shifts in the inwards direction caused by the $+ \cos(w t)$ shifts needs to be kept constant. We also notice that this forms a pattern where the y shift on the negative side of y is negative and the x-shift on the negative side of x is negative. From this, we conclude that the direction of the shift varies like a cosine function for x because the cosine function is negative in the second and third quadrant and that it varies like a sine function for y because

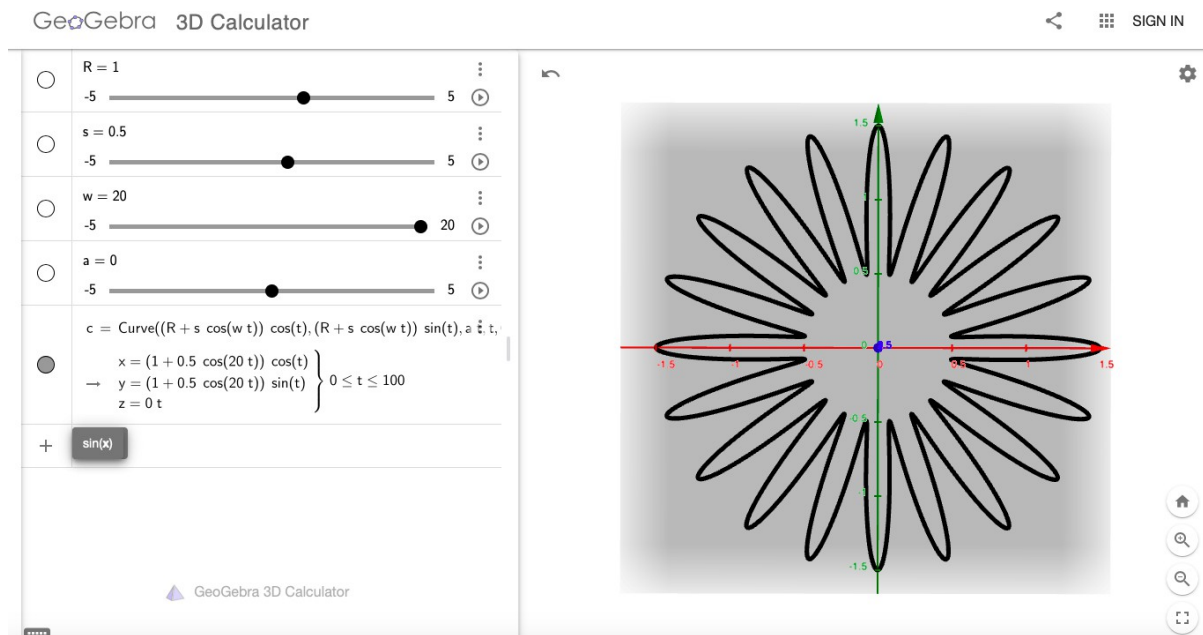
the sine function is negative in the third and fourth quadrant and positive in the first and second quadrant.

The inwards shift needs to be kept constant but the direction of the shift needs to vary circularly whereas on graph 3, it is parallel for every oscillation. As explained by the basics of cylindrical coordinates, to make a line or a shift vary circularly, it needs to be multiplied by $\cos(t)$, in the x equation and by $\sin(t)$ in the y equation. Therefore, we multiply the $+ \cos(w t)$ shift. Similar to how on curve 1, the value R is multiplied by these values in the x and y parametric equations, the same thing happens with the shift giving the following curve:

$$c_5 = \text{Curve}(R \cos(t) + s \cos(w t) \cos(t), R \sin(t) + s \cos(w t) \sin(t), a t, t, 0, 100)$$

After factorizing:

$$c_5 = \text{Curve}((R + s \cos(w t)) \cos(t), (R + s \cos(w t)) \sin(t), a t, t, 0, 100)$$



Graph 4: xy Plane Circular Cosine Wave (base curve 5)

Inputting Calculated Values for the Two-Dimensional xy Plane

Using the information that the radius of the Sun's orbit is $R = 2.5 \times 10^{20}m$, we can input this value for R. Since the Earth completes 240000000 orbits around the Sun in one orbit around the Milky Way, the circular cosine function will have 240000000 oscillations. Thus, $w = 240000000$. Furthermore, because the Semi major axis of the Ecliptic plane is $s = 1.49597888 \times 10^{11}m$ and the same minor axis is $b = 1.495770 \times 10^{11}m$, due to the

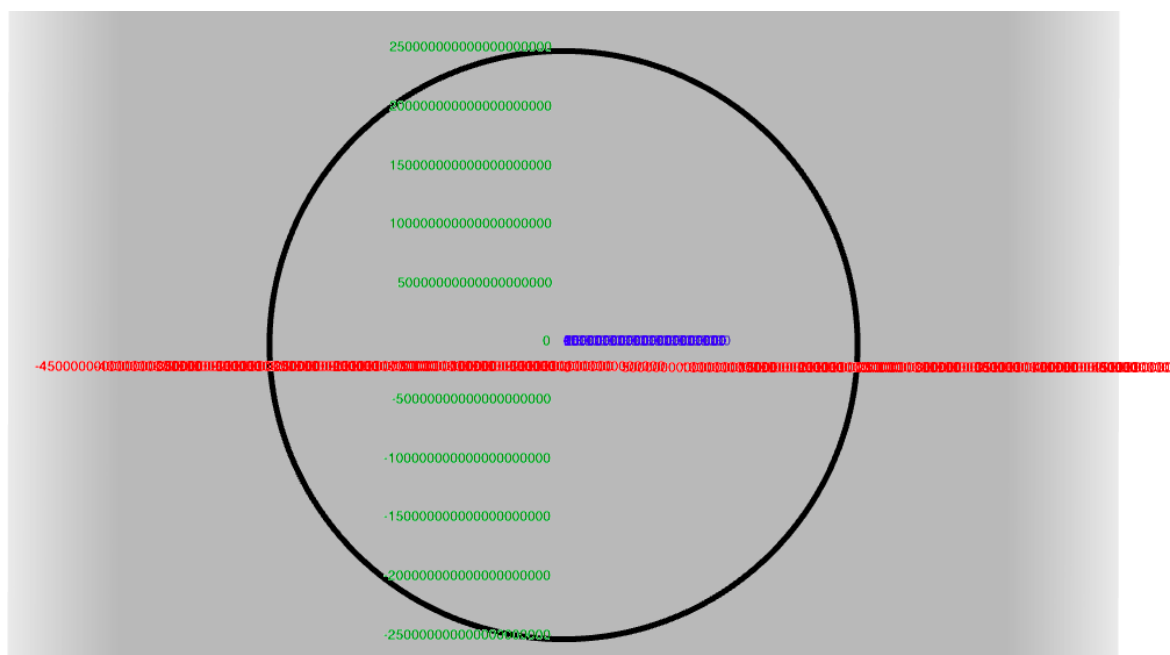
lack of information we can choose which of these stretches to use for the xy plane stretch. So let's use the semi-major axis $s = 1.49597888 \times 10^{11}m$. So:

$$s = 1.49597888 \times 10^{11} m$$

$w = 240000000$

$$R = 2.5 \times 10^{20} m.$$

Inputting these into curve 5 in Geogebra, we get the realistic 2D plane of the Earth's trajectory:



Graph 5: Realistic 2D Earth's trajectory (curve 5 with values plugged in)

This is the 2D to-scale curve of the Earth's trajectory and how the path looks viewed from above. The reason that no oscillations can be seen in the curve is because the radius of Earth's orbit around the Sun is 9 orders of magnitude smaller than the Solar System's orbit around Milky Way from the previous calculations ($s = 1.49597888 \times 10^{11}m$, $R = 2.5 \times 10^{20}m$). Furthermore, when attempting to zoom in to the curve in Geogebra to view the oscillations caused by the Earth's orbit, my computer is not powerful enough to zoom in that much and crashes before approaching the appropriate orders of magnitude.

Modelling the Third Dimension of the Base Function

To model the 3rd dimension via the z coordinate, we need to understand that when the Earth is halfway between the circular cosine peak and trough (the origin of the cosine function on diagram 2), it is at its largest displacement from the galactic plane in the z

direction because the Earth is orbiting the Sun in a circle, in 3D. Therefore, the z equation needs to reach the highest and lowest value where the circular cosine wave is at the origin value. Since the origin value is where the circular cosine wave and the circle without transformations intersect, the shift in the x and y axis here is 0. This means that z should reach the maximum and minimum value when $\cos(w t) = 0$ as that is the function for the shift. Understanding that on the basic sine and cosine graphs, sine peaks and troughs where cosine is 0, we can conclude that having z as a function of sine works for our needs.

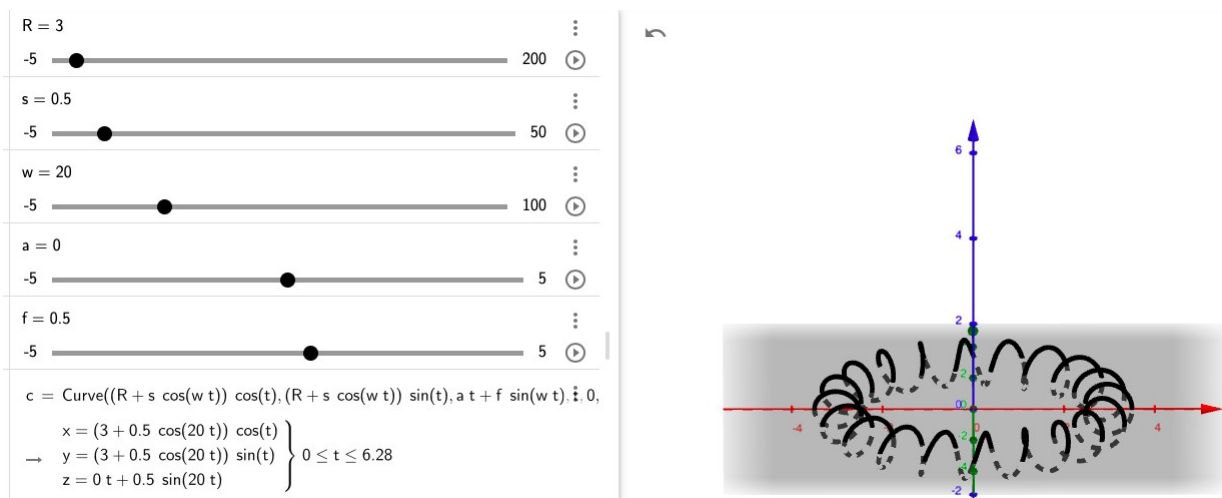
$$\Rightarrow z = \sin(t)$$

Furthermore, if for example, the function on graph 4 has 20 peaks and troughs that means that the Earth orbiting the Sun creates 20 orbits meaning that every position along that orbit needs to be reached 20 times throughout the curve. This includes the position when the z value on the 3D curve peaks. Therefore, we conclude that the number of peaks of the curve on the z -axis must match the number of periods of the circular cosine wave. To make them match, since w was used for making the circular cosine wave, the same w is used for z . We get:

$$\Rightarrow z = f \sin(wt) \quad (f \text{ is added as a factor to vary the vertical stretch of the loops})$$

Making the curve:

$$c_6 = \text{Curve}((R + s \cos(w t)) \cos(t), (R + s \cos(w t)) \sin(t), at + f \sin(wt), t, 0, 100)$$

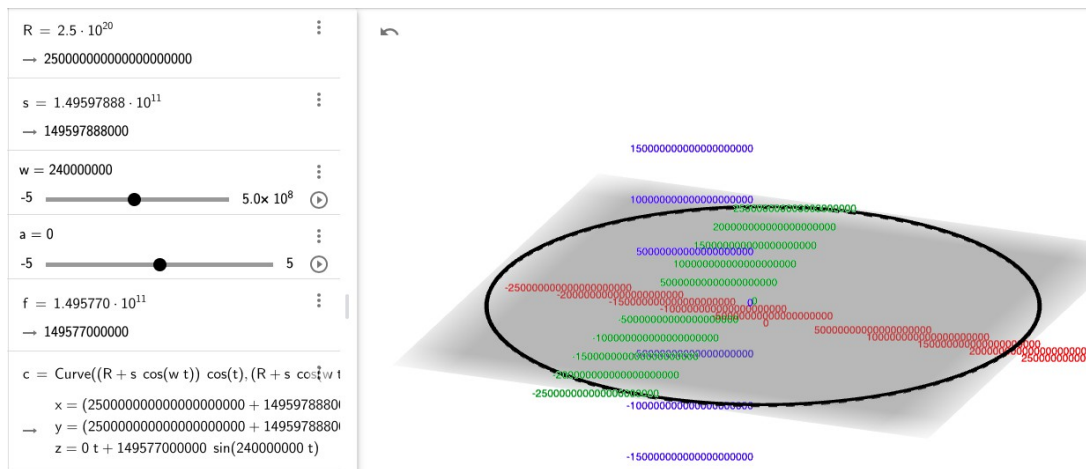


Graph 6: Circular Helix (base curve 6)

Inputting Parameters of the Earth's Ellipse

Since the semi major axis was chosen to be oriented in the xy plane, the calculated semi-minor axis stretch. ($f = 1.495770 \times 10^{11}m$) will be plugged into the function for the

vertical stretch of the loops. Calculated b can be used in the place of f because the loops on curve 6 are vertical so the vertical stretch equals the semi-minor axis:



Graph 7: Circular Helix with Plugged-in Values (curve 6 with values plugged in)

As on graph 5, we cannot see the minor helix of the Earth's orbit as it is so much smaller than the Solar System's orbit around Milky Way. Similarly to before, my computer is not powerful enough to zoom in until the oscillations caused by Earth's orbit are visible and crashes before approaching the correct orders of magnitude.

Tilting the Ecliptic Plane on the Base Function

Understanding that the ecliptic and galactic planes are not perpendicular to each other from table 1, the loops on graph 6 need to be tilted. The required shape is more clearly determined from diagram 3 (not to scale):

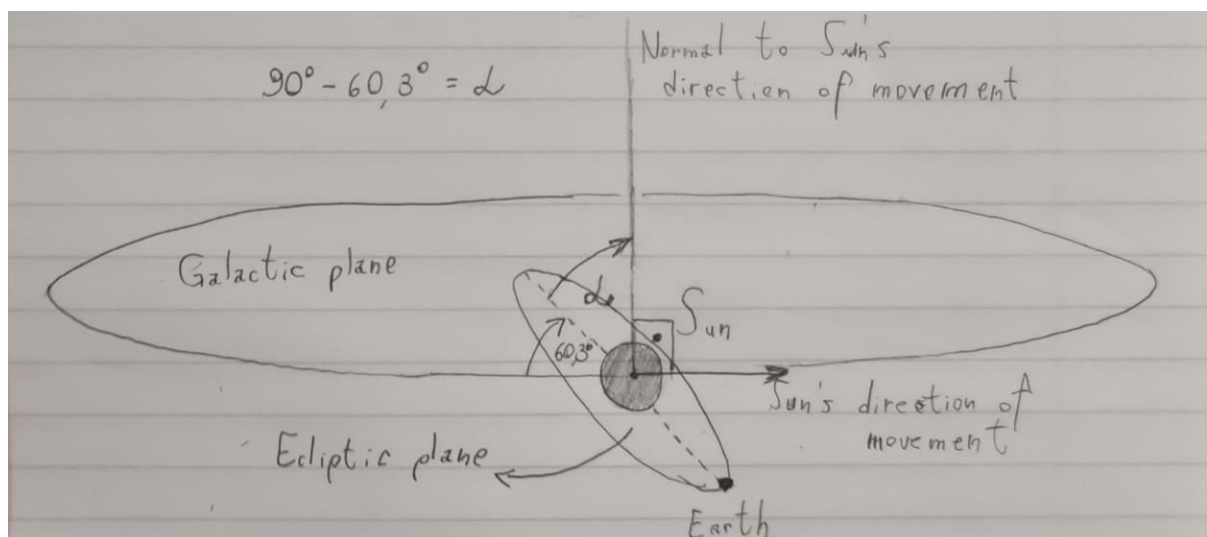


Diagram 3: Angles between the Galactic and Ecliptic Plane and Sun's Direction of Movement

Looking at the curve from graph 6, to tilt the loops along the x -axis, the x -values of the curve need to be shifted by the amount that is dependent on the height of the curve (z coordinate). Because the equation for height is given under z , we can perform this transformation by simply adding the equation of z to the x equation. When z is big or small, a big or small shift will occur in the loops respectively tilting the loops. The same thing works for the y equation. The expressions for x and y are now as follows:

$$x = (R + s \cos(w t)) \cos(t) + \sin(w t)$$

$$y = (R + s \cos(w t)) \sin(t) + \sin(w t)$$

Also, looking at the x -axis, to tilt the loops and maintain the correct orbit, the loops in the first and second quadrant need to be tilted in the opposite direction from the loops in the third and fourth quadrant. Also, the loops right above the x -axis need to have 0 shift in the x direction while the loops above the y -axis need to have maximum shift. This means that this added shift needs to vary like a sine function so it is therefore multiplied by $\sin(t)$ in the equation for x .

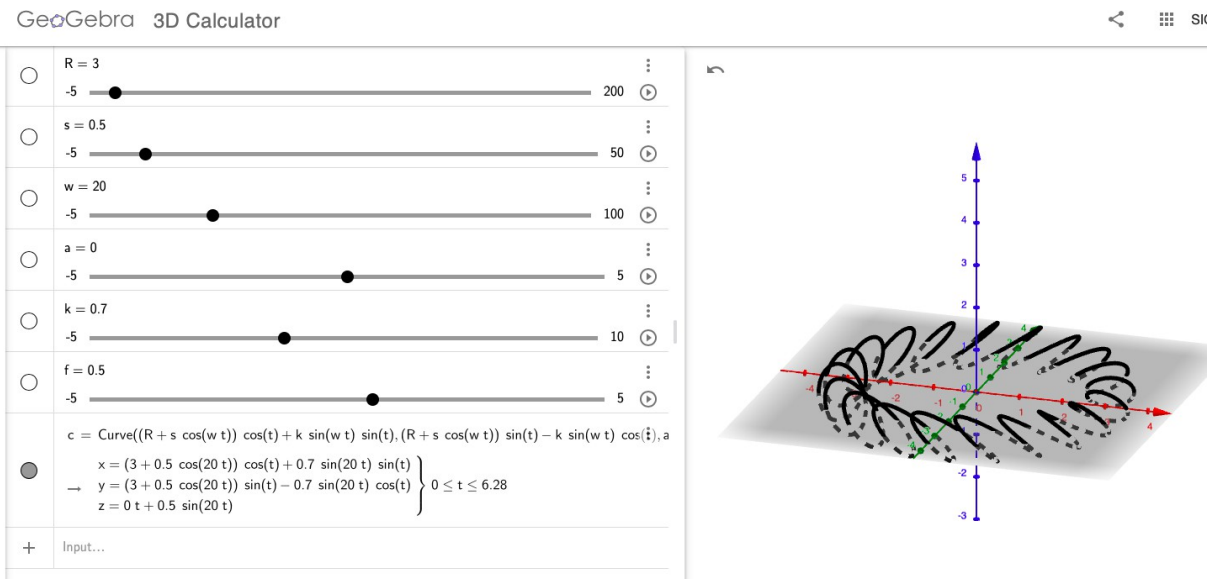
$$\Rightarrow x = (R + s \cos(w t)) \cos(t) + k \sin(w t) \sin(t)$$

Looking at the y axis, the loops in the second and third quadrant have to be shifted oppositely to the loops in the first and fourth quadrant. Also, the loops right above the y -axis need to have 0 shift in the y direction while the loops above the x -axis need to have maximum shift. This means that the shift added to y needs to vary like a cosine function. However, the value for $\cos$ is -1 above the x -axis at the border of the second and third quadrant meaning that the tilt of the loops along the y -axis will be counterclockwise opposing the tilts along the x -axis. Therefore, this tilt needs to be reversed by adding a minus to the shift for y making the tilts clockwise.

$$\Rightarrow y = (R + s \cos(w t)) \sin(t) - k \sin(w t) \cos(t)$$

With k added as a factor to vary the tilt, this gives the final base curve:

$$c_7 = \text{Curve}((R + s \cos(w t)) \cos(t) + k \sin(w t) \sin(t), (R + s \cos(w t)) \sin(t) - k \sin(w t) \cos(t), at + f \sin(wt), t, 0, 100)$$



Graph 8: The Base Function for the Earth's Trajectory (base curve 7)

This graph is a representation of the trajectory resulting from the movement displayed on diagram 3 (not to scale).

Setting the Angle of the Ecliptic Plane

Referring back to table 1, galactic and ecliptic planes are not perpendicular to each other as might be intuitive, but the angle between them is 60.3° . Therefore, amount of tilt applied to curve 7 to get the real trajectory needs to be calculated. As shown on diagram 3, if the angle between the ecliptic plane and galactic plane is 60.3° , the angle $\alpha = 29.7^\circ$ is the angle by which curve 3 needs to be tilted and not 60.3° as we might have assumed intuitively. This is because the angle between the normal to the Sun's direction of movement and its direction is already 90° .

It is difficult to determine the extent to which the function is to be tilted (the value for k that should be plugged in on the slider), because there is no single line of tilt where the trigonometric properties could be applied as the line viewed from the side varies as a sine curve. That makes perfect calculations for the value of k as the orientation of the ecliptic plane needs to be approximated. Diagram 4 shows this (not to scale):

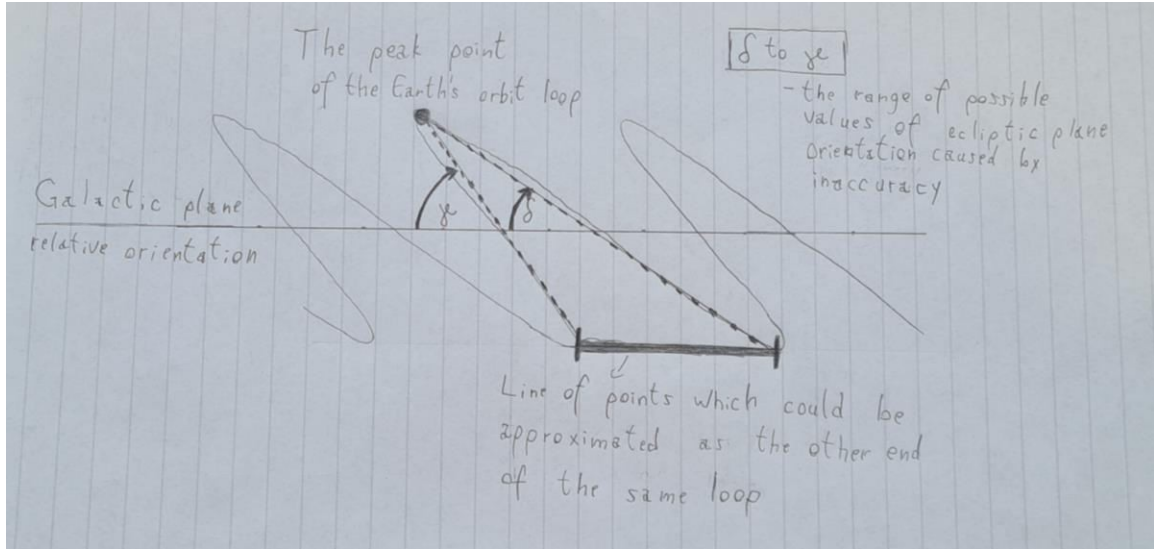


Diagram 4: Earth's trajectory viewed from the side

As viewed on diagram 4, on this curved sine shape, the current state of the ecliptic plane could be approximated for anywhere between γ and δ . This issue can be tackled by using limits. Namely, looking at the functions for x , y and z :

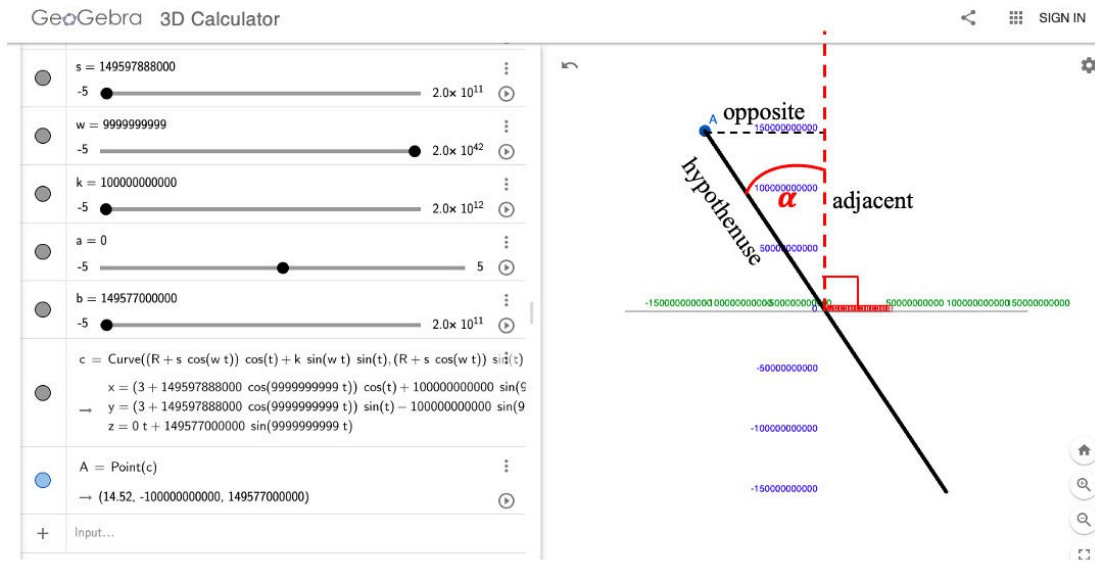
$$x = (R + s \cos(w t)) \cos(t) + k \sin(w t) \sin(t)$$

$$y = R + s \cos(w t) \sin(t) - k \sin(w t) \cos(t)$$

$$z = at + b \sin(w t)$$

Since w determines the number of oscillations as explained in “Modelling the 2D Plane of the Earth's Trajectory around Milky Way”, setting w as $\lim_{w \rightarrow \infty} c_7$ would mean that there are close to infinitely many oscillations. This would bring the loops infinitely close together. Therefore, the range of possible values for the angle between γ and δ is close to zero giving us an infinitely accurate value for that angle.

However, because there are infinitely many oscillations, the curve would look like a ring as there would be no space between each sine wave. Therefore, another limit is set for the domain of t $D_t: (0, \lim_{t \rightarrow 0})$. This limit will make the variation of t infinitely close to 0 meaning that there will be nearly zero progression of t . This will give an infinitely small number of loops which will make the curve seem like one loop. Using this loop, the value for k which gives the correct tilt of the ecliptic plane can be determined:



Graph 9: Calculating the Tilt Transformation

When viewed from the side, the loop looks like a line and since the coordinates of its peak are known with point A. Trigonometric identities are used to determine the necessary inputs of k and f by using the sine and cosine formulas to fill out the triangle. Namely, since the length of the semi minor axis is $b = 1.495770 \times 10^{11}m$, this is the desired length of the hypotenuse. The necessary adjacent side length (vertical stretch f) length can be determined by using the trig property of cosine as it relates the hypotenuse (semi-minor axis $b = 1.495770 \times 10^{11}m$) and the angle α between the ecliptic and galactic plane.

$$\Rightarrow \cos(\alpha) = \frac{f}{b} \Rightarrow f = b \cos(\alpha) \Rightarrow f = 1.495770 \times 10^{11}m \times \cos(29.7^\circ)$$

$$\Rightarrow f = 1.299273 \times 10^{11}m$$

The necessary tilt shift k (opposite side from graph 9) is calculated from the sine identity:

$$\Rightarrow \sin(\alpha) = \frac{k}{b} \Rightarrow k = b \sin(\alpha) \Rightarrow k = 1.495770 \times 10^{11}m \times \sin(29.7^\circ)$$

$$\Rightarrow k = 7.410922 \times 10^{10}m$$

Results

The base function of the Earth's orbit is as follows:

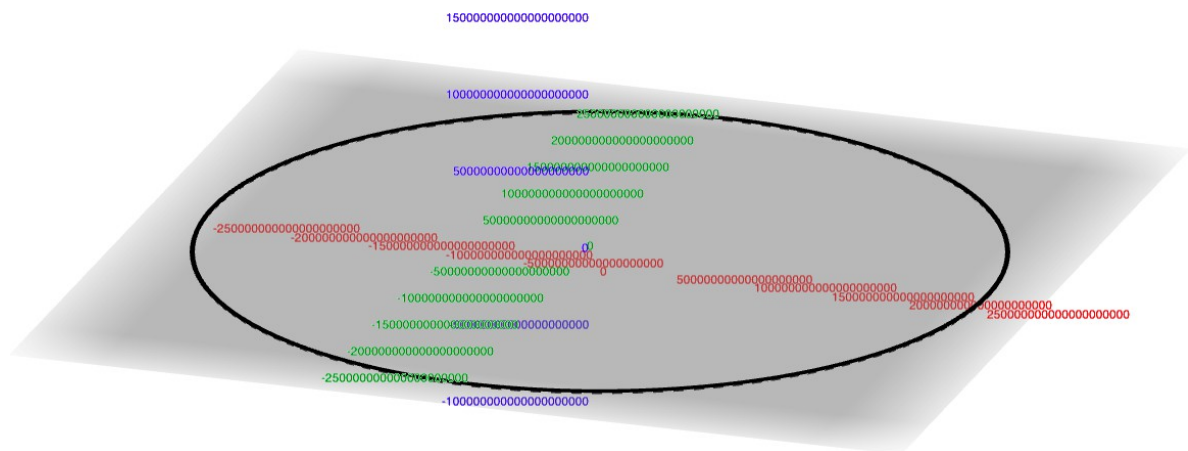
$$c_7 = \text{Curve}((R + s \cos(w t))\cos(t) + k \sin(w t)\sin(t), (R + s \cos(w t))\sin(t) - k \sin(w t)\cos(t), at + f \sin(wt), t, 0, 2\pi)$$

Paired with the following calculated values for the transformations required

Quantity	Denotation	Value
Radius of the Galactic plane	R	$2.5 \times 10^{20}m$
Ecliptic plane semi-major axis	s	$1.49597888 \times 10^{11}m$
Tilt shift of the Ecliptic plane	k	$7.410922 \times 10^{10}m$
Number of Earths orbits in a Milky Way revolution	w	240000000
Stretch of the trajectory dependent on the Milky Way velocity	a	Free to choose because of no reference point in space
Vertical stretch for giving the correct values for k and Ecliptic plane semi-minor axis	f	$1.299273 \times 10^{11}m$

Table 2: Calculated Transformations

This gives the following curve to represent the Trajectory of the Earth through the Universe



Graph 10: Final Curve with Correct Transformations

Conclusion

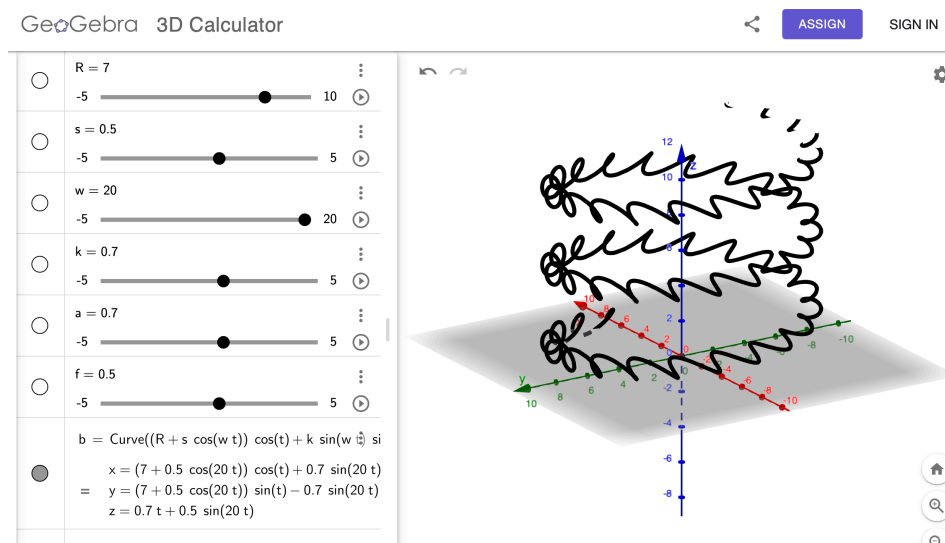
In the end, I managed to produce a curve that very accurately draws the trajectory of the Earth's movement through the universe. I combined my interest in astrophysics and transformations of functions to produce a good result using cylindrical coordinates. The aim of my investigation to model this was successfully achieved. The curve made is as accurate as it could have been with the values for universal distances I was able to access online. At the start of the investigation, I expected that creating a helix would be more difficult, however I decided to do some research to see if it can be made easier where I found cylindrical coordinates and they helped me model this. I am very happy with myself for this because I managed to contain my excitement and did not just jump into modelling. I will make sure to approach other assignments like this. During the process I figured out a lot of unconsidered factors which really explain the depth of calculations and thinking that goes into astrophysics. These are things such as whether the orbits are clockwise or counterclockwise, the difficulty of determining relative orientations for three objects which are moving circularly... This was useful as it helped me understand how complicated astrophysics is. Some findings shocked me such as that the Ecliptic plane is not perpendicular to the Galactic plane. For some reason, it was intuitive that they had to be perpendicular to go with Kepler's and Newton's Laws.

Limitations

The first error of my investigation comes from the basis of Newton's laws which state that if body 1 is acting force on body 2, body 2 has to act with same force on body 1. This means that if the Earth feels gravity from the Sun, the Sun feels gravity from the Earth and thus the Sun cannot be in a fixed position. In reality this is the case and Sun is not the center of the Earth's orbit, but the center is rather the Solar System's center of mass. This point is called the Barycenter. This makes the radius of the Earth's orbit naturally inconsistent. The distance from the barycenter could not be used as the barycenter is constantly shifting due to there being other planets in the Solar System. However, the Sun being by far the heaviest object in the Solar System, the barycenter is very close to the Sun (sometimes even inside its surface) so the error caused by not considering this is minimal. Another issue is the fact that the Moon throws off the Earth's trajectory by a little with its gravitational pull. After learning more about how the physics of this work out, it would be nice to do an extension where the barycenter is taken into account. The other significant drawback to my investigation is

not anticipating that the relative radius of the Earth's orbit compared to the radius to the orbit of the Sun is so small that my computer is not strong enough to zoom into the curve to show the helix. When trying to solve this and getting crashes on my computer, I was very disappointed as I was making this curve with excitement as to how I am going to be able to zoom in and trace the path of the Earth. It would be nice to maybe plug in this curve to a more powerful computer or to find a program stronger than Geogebra.

Furthermore, there are many perspectives as to the movement of the Milky Way to the Galaxy. Some scientists think that we should view its relative motion to the constellations of Leo and Virgo¹⁰ thus making the stretch “a” a calculatable value. Some would argue that reference zero velocity is the average velocity of the galaxies surrounding Milky Way. As there are many perspectives to this issue, this may also be viewed as a limitation to the investigation. I did not want to take sides on this argument as my knowledge is not advanced enough to argue against world-renowned astrophysicists so I decided to leave this stretch as a slider. If you wish to also model the movement of the Earth through the universe which would mean that the black hole at the center of Milky Way's is no longer our reference point meaning it is not fixed in space and its velocity is not zero, you can apply the ‘a’ stretch and the path would look like this:



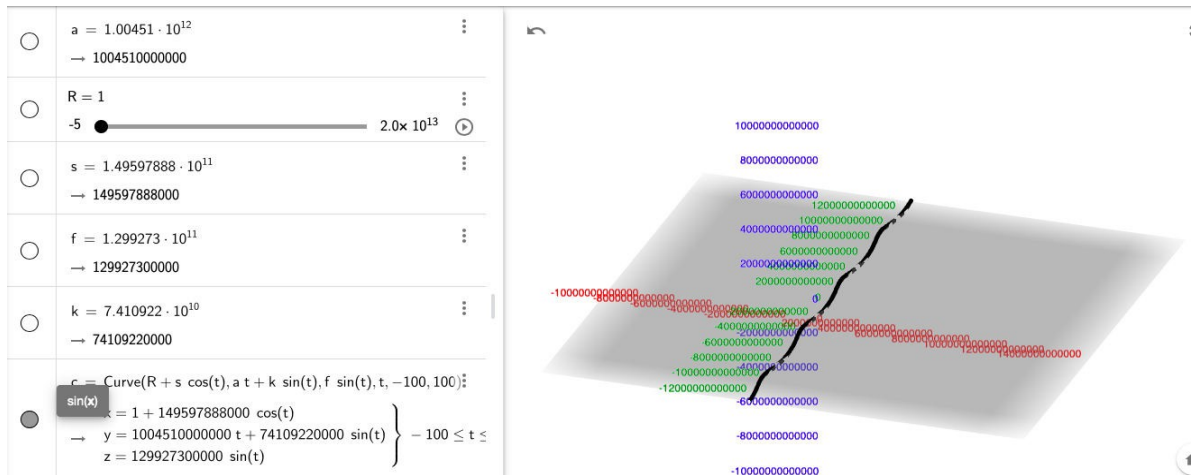
Graph 11: Earth's Path through the Universe assuming a reference point other than the center of Milky Way

Now, the function represents 3 key movements (in order from more zoomed in to more general):

4. Earth's orbit around the Sun (the small, tilted loops)
5. Movement of the Sun through the Milky way (the circle that the small loops travel in)
6. The movement of Milky Way through the Galaxy (Slow vertically upward progression of the small loops in a loop, forming a double helix on graph 11)

Closeup of the Trajectory

Wishing badly to see how the trajectory looks zoomed in, I made an extra function that represents this movement. It was made large enough by not considering the distance of the Sun from the Milky Way center. Rather I just plugged in the radius of the helix produced from the Earth orbiting the Sun as it moves through Milky Way and the tilt of the Ecliptic plane. This is how the trajectory would look if I had a strong enough computer to zoom in enough:



Graph 12: Zoomed in Trajectory of the Earth

¹⁰ (NASA 2007)

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